

THE AURAX

WhatsApp AI Sales Bot — End-to-End Test Report

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EXECUTIVE SUMMARY

The bot's own software — conversation engine, sales funnel, admin console, and safety fallbacks — is working correctly and was proven with live calls. The failures found are all EXTERNAL service/config issues (the WooCommerce store's API, the Groq provider, MongoDB/Sheets credentials), not bot code defects. The bot degrades gracefully around every one of them.

6 PASS 2 FAIL 3 DEGRADED (graceful fallback active)

1. Feature Scorecard

Core purchase funnel (search !' size !' qty !' address !' confirm !' invoice)	PASS
Deterministic fast paths (FAQ, size/qty, confirm, language)	PASS
Knowledge Hub (teach !' confident zero-LLM answer)	PASS
Admin console + API auth + all monitoring endpoints	PASS
LLM fallback chain (survives Groq failure via Fireworks/Sarvam)	PASS
Invoice PDF generation + graceful WooCommerce fallback	PASS
WooCommerce live API (orders, tracking, product sync)	FAIL
Groq primary provider tool-calling	FAIL
MongoDB persistence (falls back to JSON)	DEGRADED
Google Sheets lead logging	DEGRADED
Telegram owner alerts (not configured)	DEGRADED

2. Findings — ranked by severity

CRITICAL WooCommerce live API returns 401 — orders, tracking & product sync are down

Authenticated wc/v3 calls (with the consumer key/secret) are rejected with a NON-standard body: {"success":false,"message":"API is working, Site Connected"}. The public REST root and Store API return 200, so the site is up — a plugin ("Site Connected", an app-connector/security plugin) is intercepting authenticated admin-API requests. Impact: (a) real order creation on "confirm" fails — bot falls back to a PDF invoice; (b) order TRACKING (lookup_order) has NO fallback and returns an error to the customer; (c) npm run sync cannot refresh products (bot serves a cached list from 11-Jul).

Fix: In WordPress admin, identify the plugin returning "Site Connected" (app builder / security / "1cc" checkout connector) and whitelist the WooCommerce REST API, or regenerate the API keys under WooCommerce !' Settings !' Advanced !' REST API and confirm the key's user has the Shop Manager role. Verify with: GET /wp-json/wc/v3/orders?per_page=1 returning 200.

HIGH Groq (primary LLM) tool-calling failing on all 5 keys — not transient

Every Groq key returned 400 "Failed to call a function / tool_use_failed" on every attempt during live tests, so all traffic fell through to the paid Fireworks/Sarvam providers. The bot still answered correctly, but latency is higher ("H20 failed attempts per turn before fallback) and the free primary provider is effectively bypassed, shifting load onto paid providers.

Fix: Re-verify the Groq model id (llama-3.3-70b-versatile may be deprecated/renamed on Groq) and test a bare tool-call; if the model changed, update GROQ_MODEL. Until fixed, Fireworks/Sarvam correctly carry the load.

MEDIUM MongoDB connection fails (TLS handshake) — silently uses JSON files

Despite MONGODB_URI being set, the driver throws an SSL/TLS "internal error" (alert 80) and the bot falls back to local JSON stores. All persistence still works, but it is single-file JSON, not the intended MongoDB.

Fix: Check the Atlas cluster TLS/IP-allowlist and Node/OpenSSL version; confirm the SRV string and that this machine's IP is allowlisted. The JSON fallback is safe in the meantime.

MEDIUM Google Sheets lead logging denied — "caller does not have permission"

During live lead logging the Sheets API rejected the write with a permission error. First-contact leads are NOT reaching the spreadsheet (they are still saved to the local leads/customers stores, so no lead is lost).

Fix: Share the target Google Sheet (GOOGLE_SHEETS_ID) with the service-account email found in credentials.json, granting Editor access.

LOW Telegram owner alerts not configured

TELEGRAM_BOT_TOKEN / TELEGRAM_CHAT_ID are unset, so bulk-order/escalation alerts via Telegram will not fire. Owner alerts via WhatsApp still work; this is an optional secondary channel.

Fix: Set the two env vars if a Telegram alert channel is wanted; otherwise no action needed.

LOW Deep fallbacks degraded: OpenAI key out of quota (429), OpenRouter free model deprecated (404)

These are the last-resort providers below Groq/Fireworks/Sarvam. With the top three healthy they are never reached, but they no longer provide a safety net.

Fix: Top up/replace the OpenAI key and update OPENROUTER_MODEL to a currently-available slug if deep redundancy is desired.

LOW Minor: mid-flow quantity change leaked raw tool JSON

In the funnel, replying "2 pieces" AFTER the item was already added returned the raw tool payload (`{\"productId\":\"...\",\"qty\":2}`) as the visible message and the quantity did not update (order summary still showed qty 1). Rare edge case; the normal size+qty-in-one-message path works correctly.

Fix: In answerQuery, guard against emitting a tool-call arguments object as replyText on a quantity-only update turn.

3. Verified working (live proof)

'Full sales funnel — IDLE !' COLLECTING_ADDRESS !' CONFIRMING_ORDER !' IDLE all transitioned correctly; order confirmed and a proforma invoice URL was issued.

'Graceful WooCommerce fallback — When createOrder hit the 401, the bot automatically generated a PDF proforma invoice instead of erroring to the customer.

'Language detection — 5/5 correct — English vs Tanglish decided deterministically (incl. Tamil-script and romanized "enna price da").

'Deterministic fast paths — "M size 2" !' {size:M, qty:2}; pure "YES" !' deterministic_confirm; non-parseable text !' null (falls to LLM). Zero-LLM as designed.

'FAQ matcher — COD and shipping-time questions answered instantly from faq.json.

'Knowledge Hub — A seeded "warranty" correction was served directly (intent=knowledge, zero LLM) through the full answerQuery path.

'Tanglish provider (Sarvam) — sarvam-30b answered in natural Tanglish ("Semma choice bro!") with tool-calling, 121 output tokens — cheap & correct.

'Invoice PDF — Valid %PDF file generated (24.8 KB) with correct header.

'Admin console + auth — SPA served at /admin; unauthenticated API = 401; wrong password = 401; correct password issues a bearer token; all 9 monitoring endpoints return 200.

'Conversation review / diagnose — Review flagged 5 of 70 leads; auto-diagnose queue operational.

'Persistence (JSON) — Sessions, leads, customers, and retry-queue read/write verified.

4. Recommended action order

1. Fix the WooCommerce API 401 (whitelist REST API / regenerate keys) — restores order creation, live order tracking, and product sync. Highest customer impact.
2. Fix Groq tool-calling (verify/adjust GROQ_MODEL) — returns the cheap primary provider to service and cuts latency.
3. Restore MongoDB TLS and share the Google Sheet with the service account — both currently degrade silently but lose no data.

4. Optional: configure Telegram, refresh OpenAI/OpenRouter fallbacks, and patch the mid-flow qty-change edge case.

Evidence files retained in the repo: e2e_results_part1.json (deterministic + providers), e2e_results_part2.json (live agentic funnel), e2e_results_part3.json (server endpoints). This report was produced from live test runs against the production configuration on 2026-07-26.